

p i p e r

A custom TV interface for Raspberry Pi — a radial dial that wraps Netflix, YouTube, Disney+, Prime Video, HBO Max, Plex, Kodi, local media and any web address, driven entirely by a 5-way remote.

interaction model

D-pad spins the ring

▲▼ crosses rings

OK launches · BACK returns

dial logic

9 slots per ring

overflow creates a new ring

search always holds slot 00

honesty rule

piper logs launch time and session length only — never what happens inside an app

interactive prototype · 13 states captured from the live build · September 2026

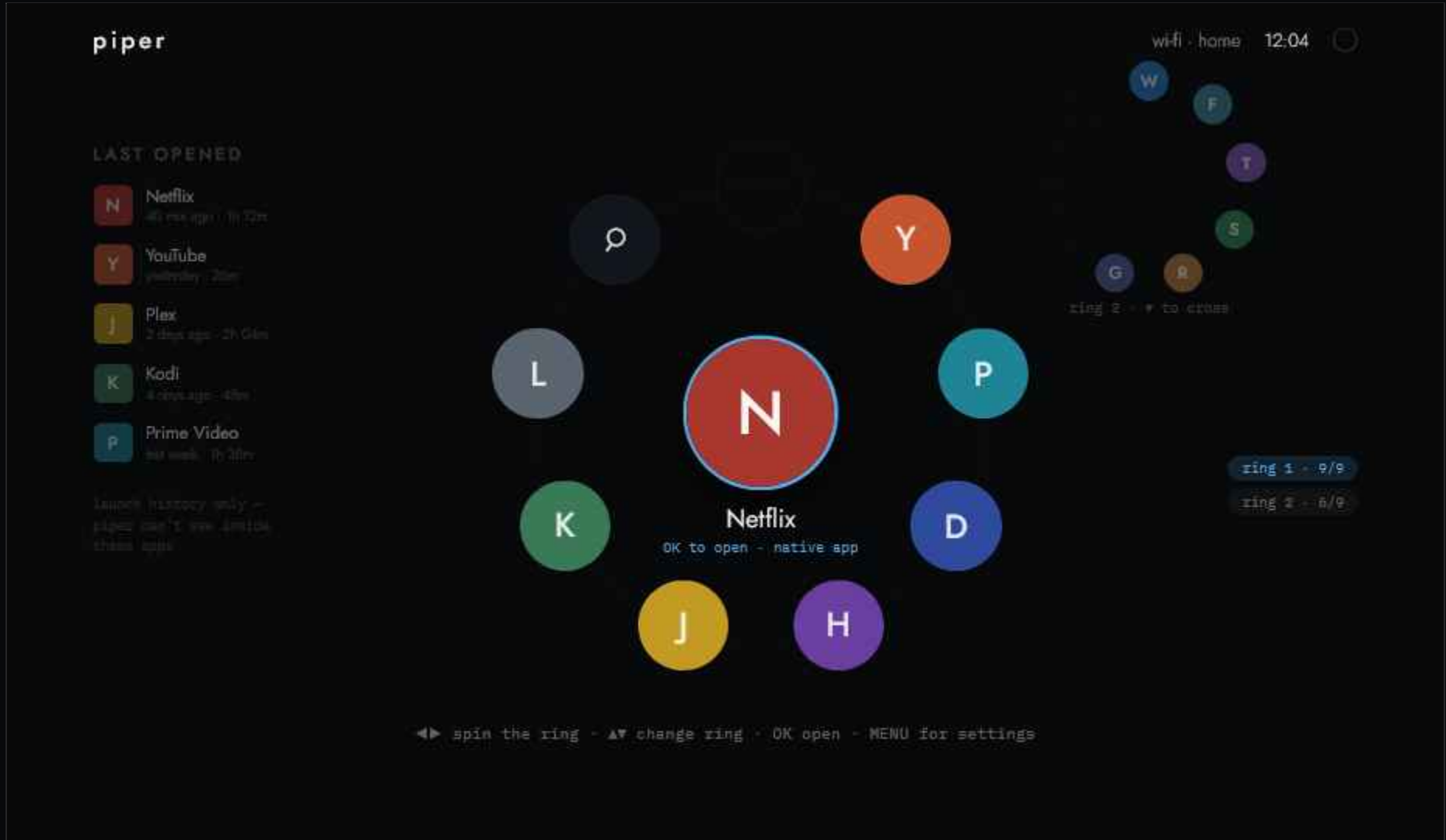
p i p e r

hdmi 2 active • pairing remote • 14 services ready

raspberry pi • T2.04 • with home

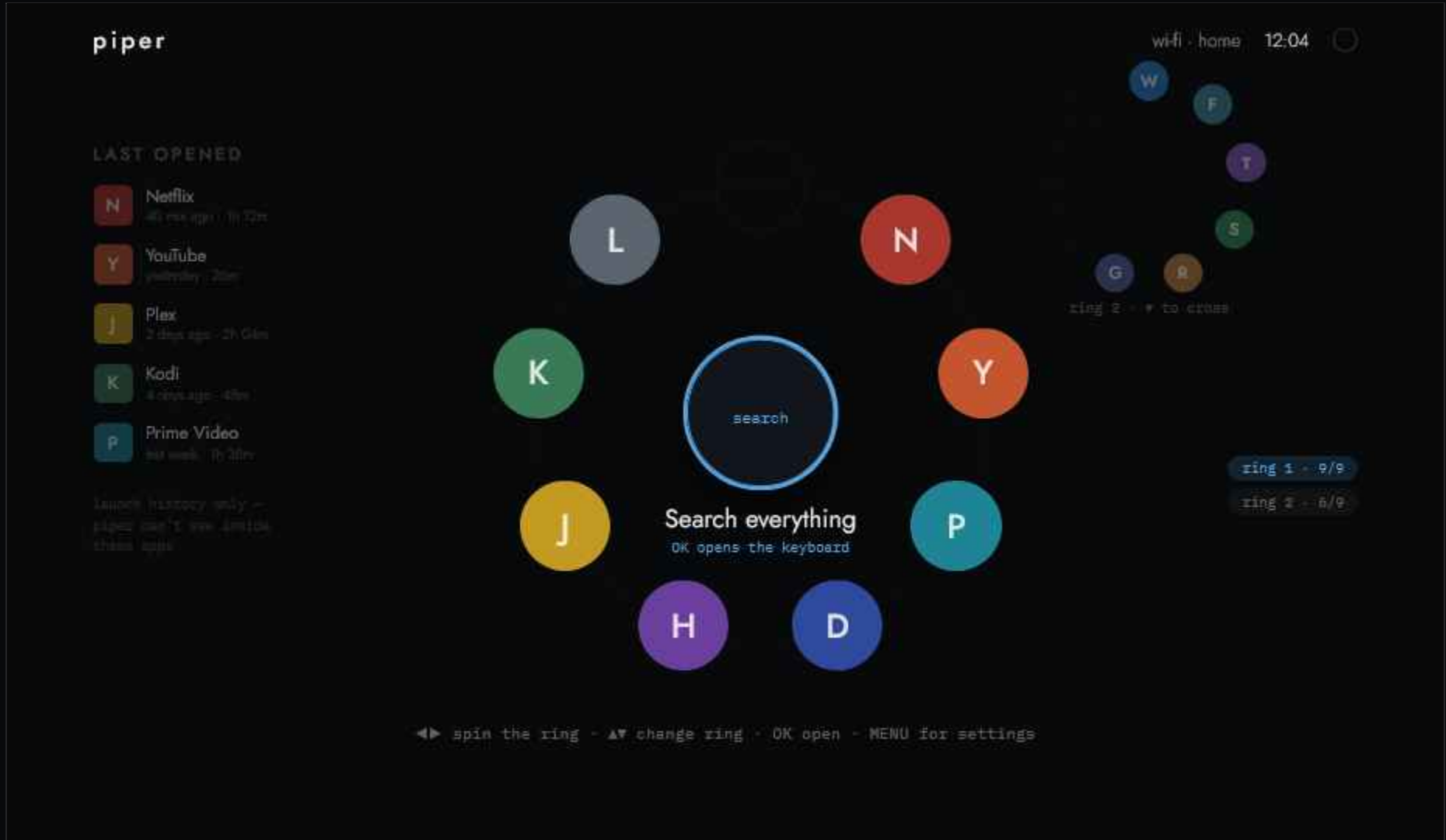
The Pi claims the HDMI input and pairs the remote before anything else appears. The service count is read from the mapping file, so the dial is already built by the time the progress bar finishes.

```
hdmi 2 active
pairing remote
14 services ready
```



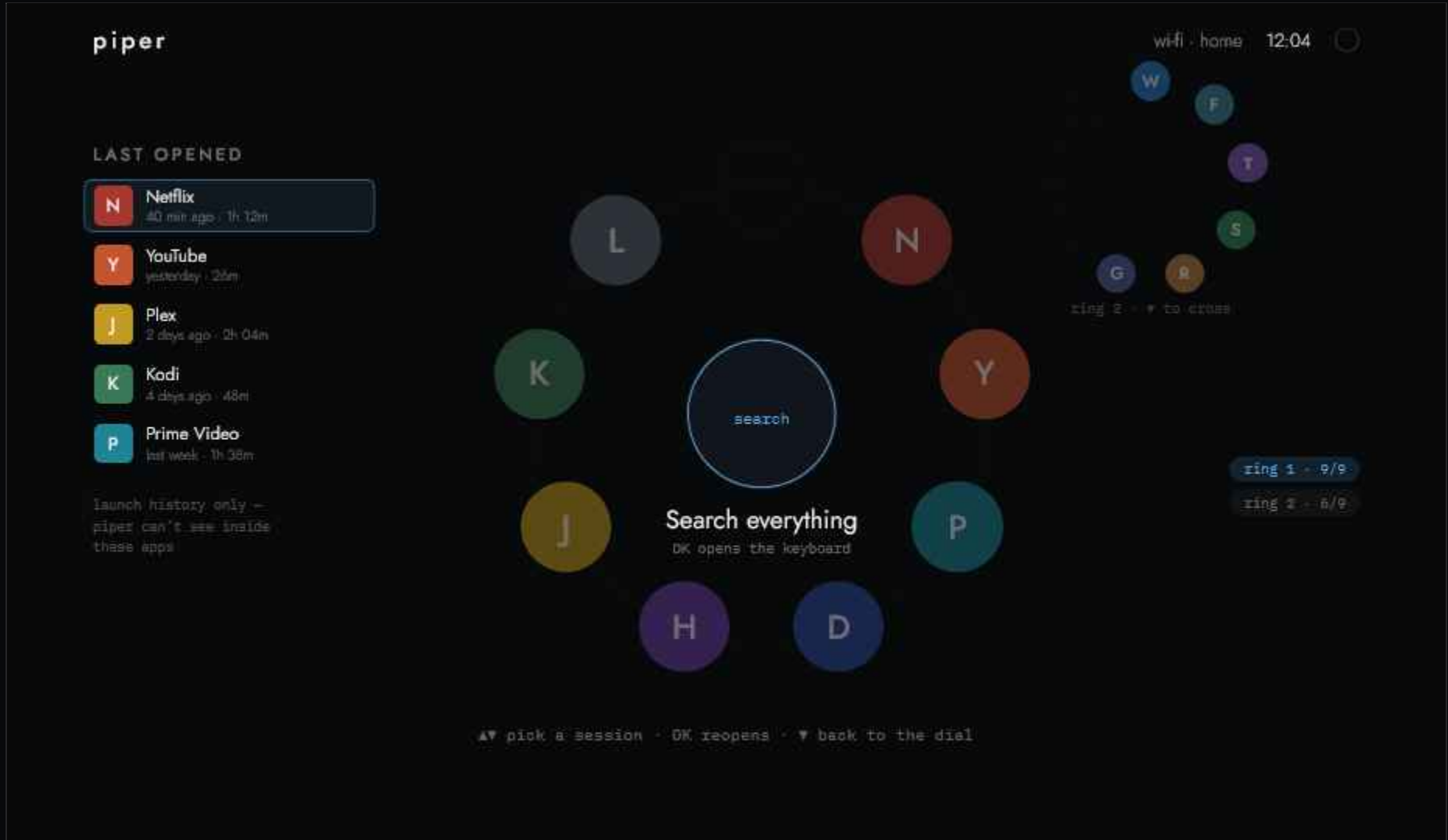
The focused service lifts into the centre of the ring at full size; everything else orbits it. Rotating with ◀▶ turns the whole ring so the focus position never moves — the eye stays in one place.

ring 1 · 9/9
slot 01 focused
Netflix · native app



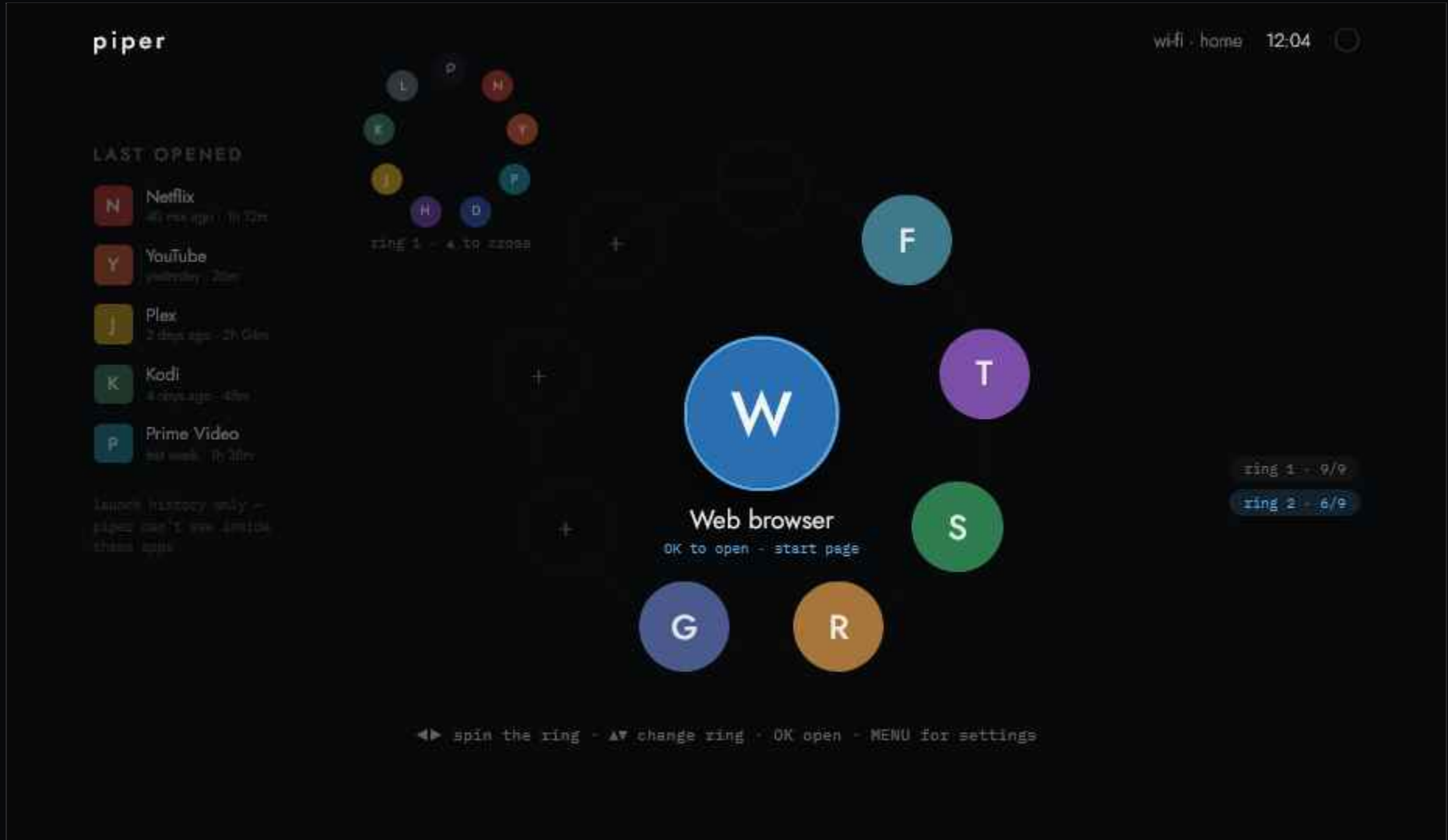
Slot 00 is always search, so it is one press away from the default focus. It is the only slot that is not a service, and it is pinned across every ring configuration.

slot 00 · reserved
OK opens the keyboard



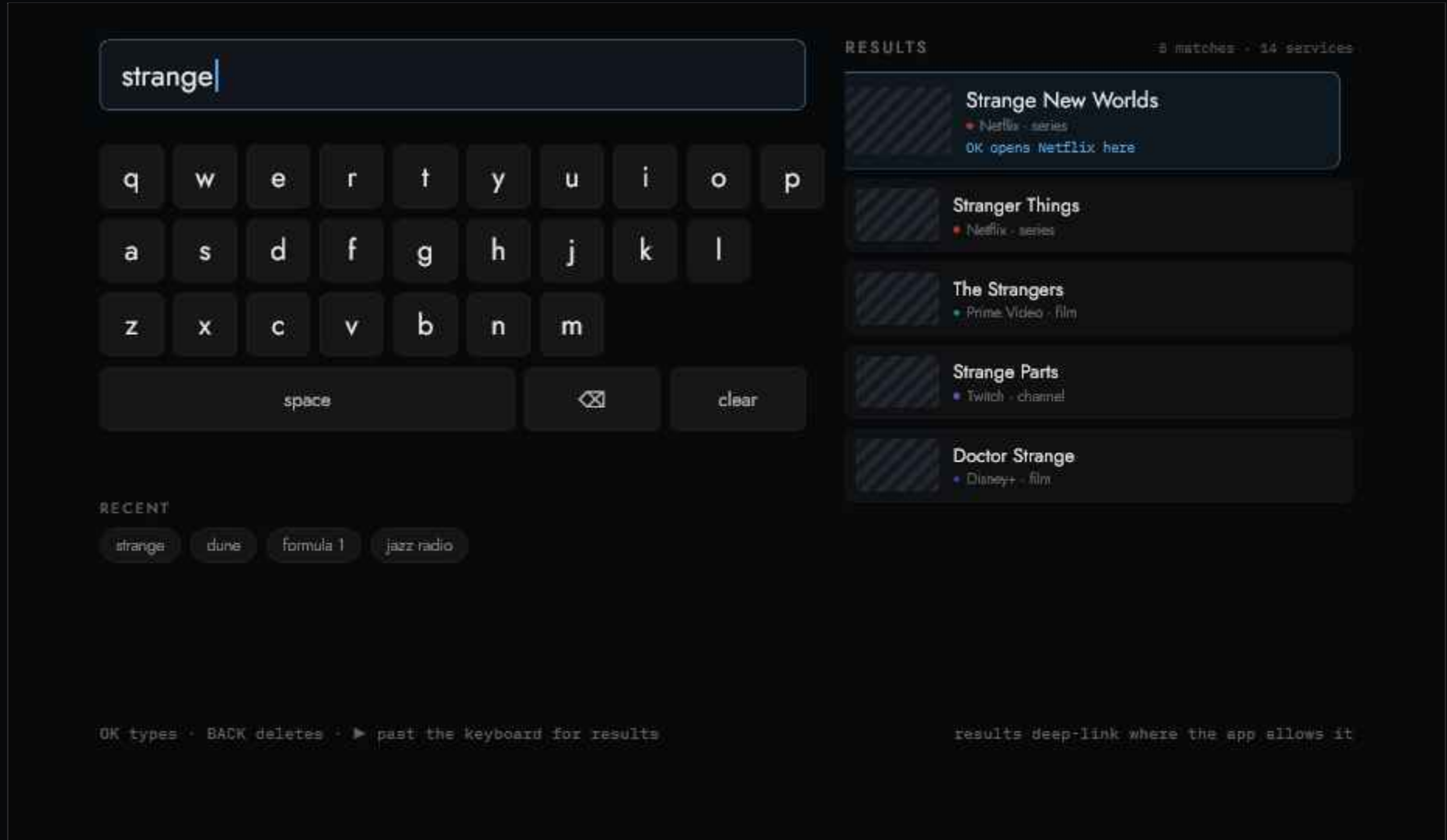
Instead of a fake “continue watching” row, piper shows only what it can honestly know: when each service was launched and how long the session lasted. Pressing OK reopens that service.

launch history only –
piper cannot see inside
these apps



Rings hold nine slots. Crossing to ring 2 keeps the previous ring visible as a dimmed mini-wheel on the left and the next one on the right, so the user always knows where they are in the stack.

ring 2 · 5/9
previous ring shown left
ring pager right



An on-screen QWERTY filters as you type. The right column only lists services that expose a search API — results deep-link into the app where the platform allows it, and say so plainly where it does not.

query "strange"
5 matches
▶ crosses to the results

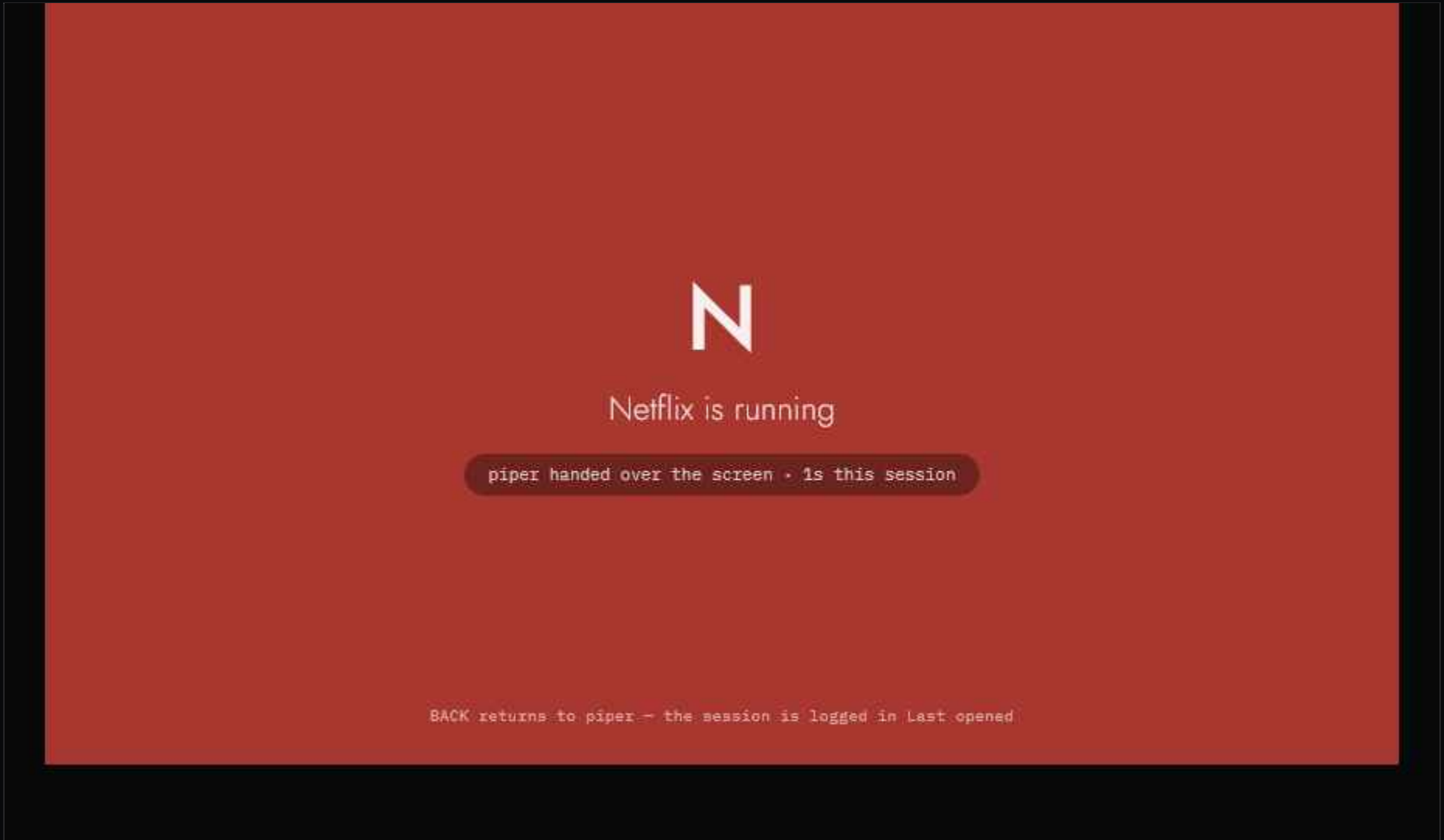


Starting Netflix

native app • remote stays in focus mode
BACK returns to piper, the app keeps running

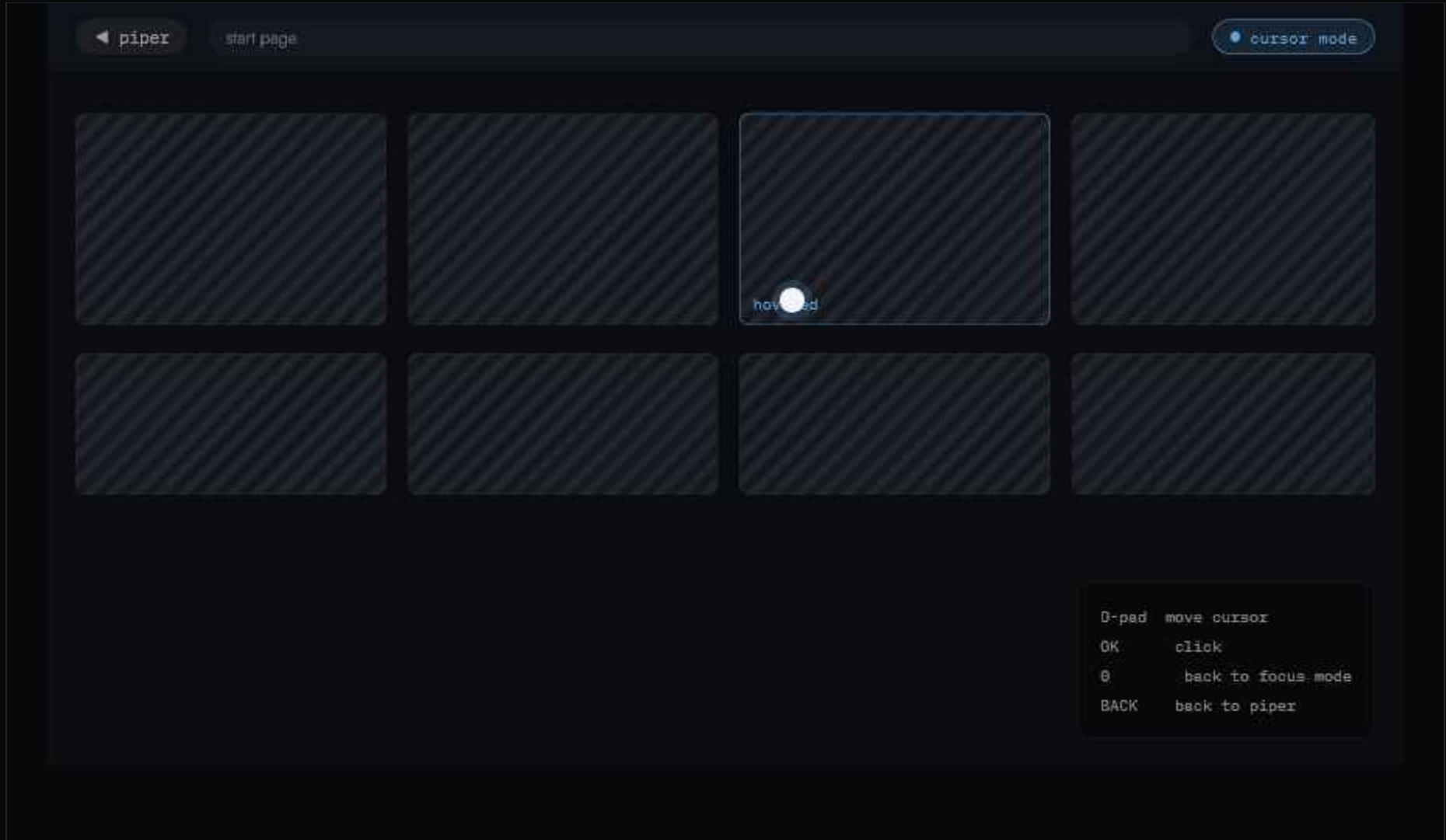
A deliberate hand-off screen. It names the launch type — native app or web app — and states up front that BACK returns to piper while the app keeps running.

native app
remote stays in focus mode



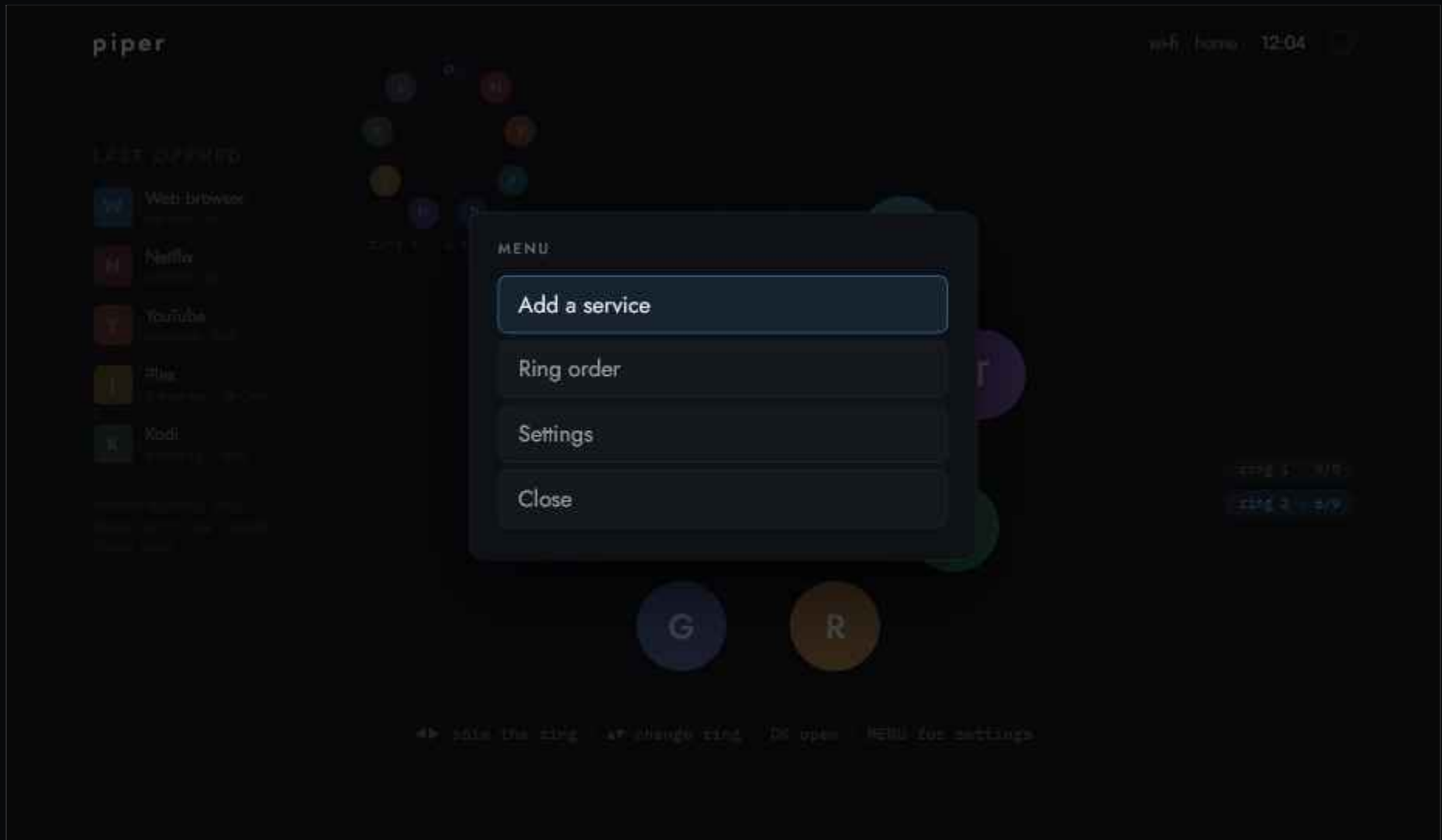
piper steps aside completely for a native app; the only thing it keeps is a session timer, which becomes the Last opened entry when the user presses BACK.

screen handed over
session timer running
BACK returns to piper



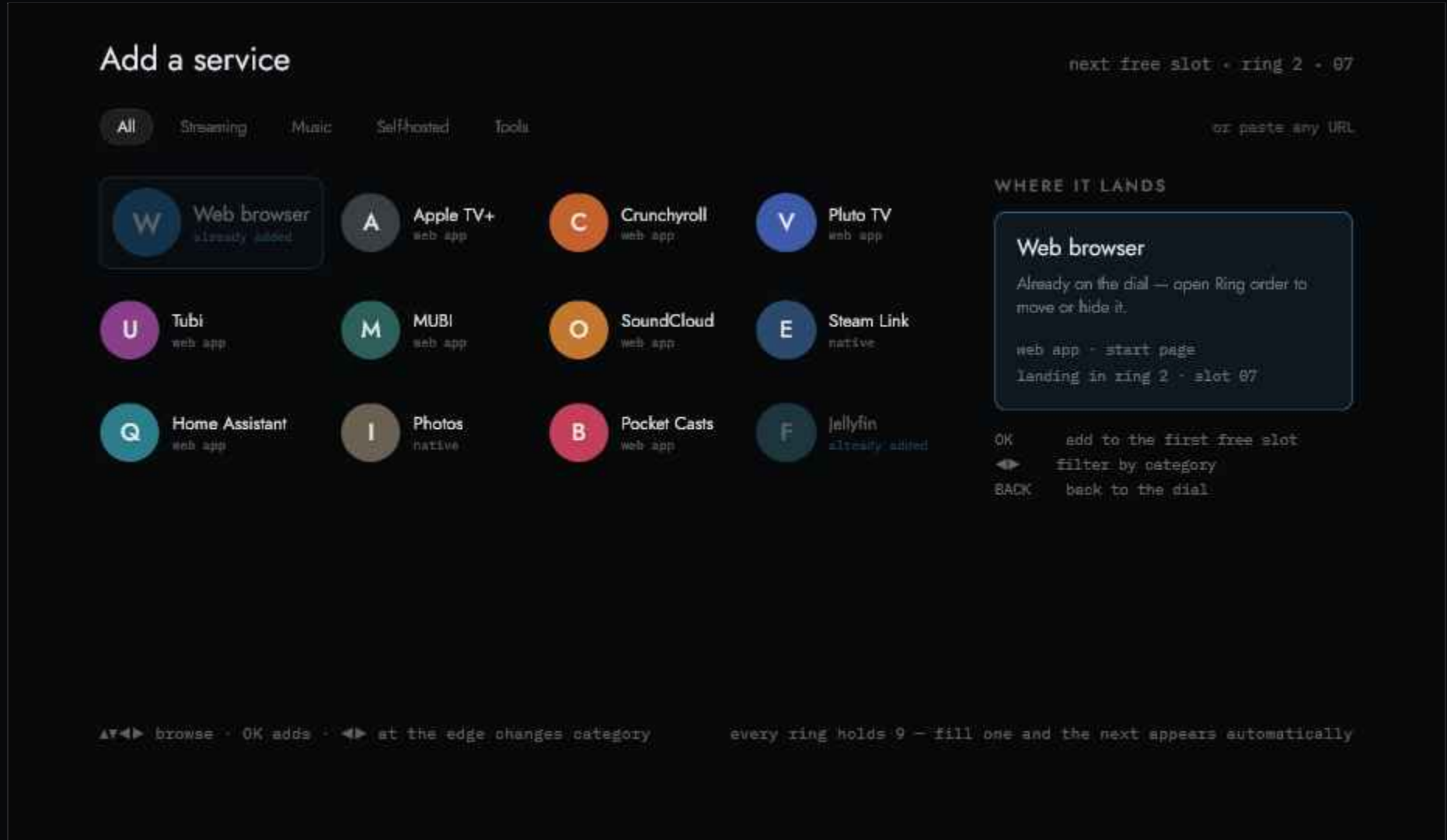
Web services get browser chrome and a pointer the D-pad drives. Tiles highlight under the cursor and OK clicks, which makes sites that were never designed for a remote usable without a mouse.

D-pad moves the cursor
OK clicks
0 back to focus mode



One overlay covers everything structural: adding a service, reordering the rings, and settings. It is reachable from any screen so the user never hunts for configuration.

```
add a service
ring order
settings
```



A catalog of ready-made entries plus a paste-any-URL path. Already-added services are dimmed rather than hidden, and the panel on the right states exactly which ring and slot the new entry will land in — creating a ring if the current ones are full.

category chips
already-added dimmed
landing slot shown

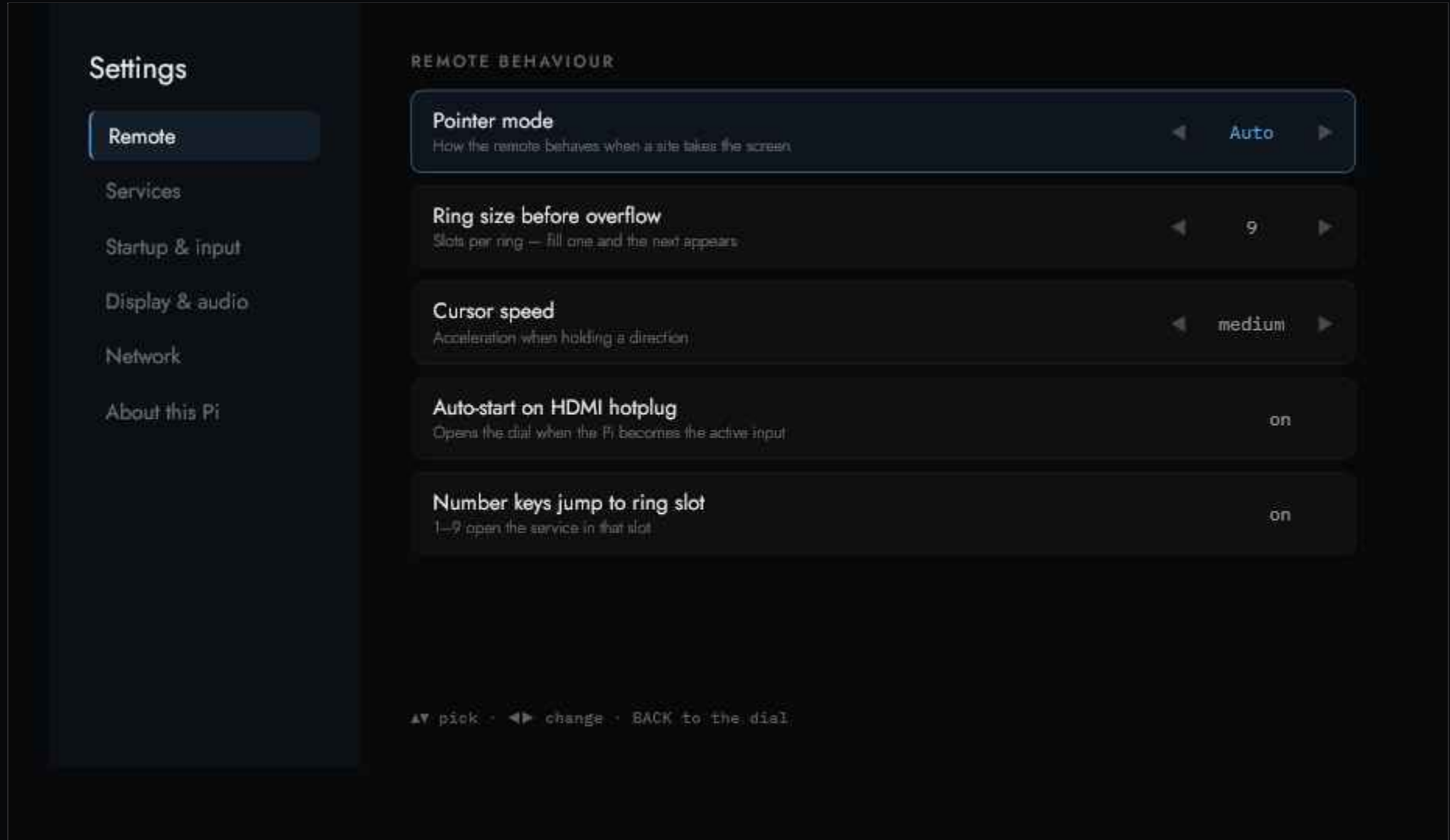
Ring order 14 of 14 on the dial · 2 rings of 9

r1 02	≡	 Netflix	just now 
r1 03	≡	 YouTube	yesterday 
r1 04	≡	 Prime Video	never opened 
r1 05	≡	 Disney+	never opened 
r1 06	≡	 HBO Max	never opened 
r1 07	≡	 Plex	2 days ago 
r1 08	≡	 Kodi	4 days ago 
r1 09	≡	 Local media	never opened 
r2	≡	 Web browser	just now 

OK to grab and reorder · ◀▶ hides a service · BACK to the dial toggle = on the ring

Every service in one list with its ring and slot number. Grabbing a row and moving it re-lays the dial immediately; the toggle takes a service off the ring without deleting it, and the slot numbers renumber live.

r1 · 01 ... r2 · 05
toggle = on the ring
never-opened shown honestly



Remote behaviour first, because it is the part of a TV interface people actually tune. Ring size before overflow is the structural one — changing it re-splits every ring on the spot.

```
pointer mode
ring size 6-12
cursor speed
```